PROJECT PROPOSAL

MOBILE APPLICATION DEVELOPEMENT

SUBMITTED TO: JAWAD KHAN

MUHAMMAD ABDULLAH (FA23-BCS-051)
ABDULAHAD (FA23-BCS-005)
ABDUREHMAN JADOON (FA23-BCS-006)

VYBIN: A Mobile Chat Application

1. Introduction

Chat apps like WhatsApp are used by millions of people every day to send messages, share photos, and talk to friends and family. For this lab project, we propose to build a simple chat application called VYBIN. The app will work on both Android and iOS phones and will let users send messages to each other in real time.

The app will be built using Flutter, a framework that allows us to write one set of code for both Android and iOS. For storing and sending data, we will use Firebase, a service from Google that helps apps save data and send messages quickly.

2. Problem Statement

Many students want to learn how real chat apps work, but most tutorials only cover small parts of the app, such as the login page or a single chat screen. There is a need for a complete, working example that covers the full process: user login, sending messages, sharing files, and keeping the app secure. This project aims to fill that gap by building a full chat app from start to finish.

3. Objectives

- Build a mobile chat app that works on both Android and iOS
- Allow users to create an account and log in safely
- Let users send and receive text messages in real time
- Allow sharing of photos, voice messages, and documents
- Keep user data and chats private and secure
- Learn how to use Flutter and Firebase together in a real project

4. Scope of the Project

This project will focus on building the core parts of a chat app. It will not try to copy every feature of WhatsApp. The first version of the app will include one-on-one chats only. Group chats and video calls are listed as future improvements and are not part of the main scope.

5. Key Features

- Real-time messaging: Messages appear on the other person's phone right away
- Media sharing: Users can send images, voice notes, and documents
- Push notifications: Users get notified when they receive a new message
- Offline access: Users can still see old chats even without internet
- Simple and clean design: Easy to use for anyone, even first-time users
- Privacy: Messages are removed from the server once they reach the receiver

6. Tools and Technologies

- Flutter and Dart – for building the app screens and logic
- Firebase Firestore – for sending and storing messages
- Firebase Authentication – for user login and sign up
- Android Studio / VS Code – as the code editor
- Git and GitHub – for keeping track of code changes

7. System Overview

The app has two main parts: the front end and the back end. The front end is what the user sees and touches, such as the chat screen and the contact list. This part is built using Flutter.

The back end handles the data. When a user sends a message, the app sends it to Firebase Firestore. Firestore then delivers the message to the other user's phone. Once the message is received, it is deleted from Firestore, so it is not stored anywhere except on the two phones involved in the chat.

8. Expected Outcome

By the end of this lab project, we expect to have a working chat app installed on a real or virtual phone. The app should let two users sign up, log in, and send messages, photos, and voice notes to each other in real time. This will show a clear understanding of mobile app development and how to connect an app to a live database.

9. Project Timeline

The table below shows a simple plan for finishing the project in seven weeks.

Week	Task
1	Study the project idea and plan the app screens
2	Set up Flutter project and connect Firebase
3	Build login, sign up, and contact list screens
4	Build the chat screen and real-time messaging
5	Add media sharing (images, voice, documents)
6	Add push notifications and offline support
7	Testing, bug fixing, and final report

10. Conclusion

This project will give us hands-on experience in building a complete mobile application from scratch. By the end, we will understand how real-time chat apps work behind the scenes, and we will have a working app that we built and tested ourselves.